

## 1. Understanding the concepts

Mark the correct statements by  $\blacksquare$  and the incorrect ones by  $\square$ . You learn from your mistakes.

- ☐ A topology may contain sets that are neither open nor closed.
- ☐ Choosing all subsets of a set  $M$  defines a particular topology.
- ☐ A set with two elements can be equipped with exactly four different topologies.
- ☐ Any map  $\varphi : M \rightarrow N$  is continuous if the topology on the domain  $M$  contains all subsets of  $M$ .
- ☐ Any map  $\varphi : M \rightarrow N$  is continuous if the topology on the target space  $N$  is the set  $\{\emptyset, N\}$ .
- ☐ The only consistent choice for a topology on the real numbers is the standard topology.
- ☐ An arbitrary set  $M$  can be equipped with at least two more topologies than the standard topology.
- ☐ There always exist enough charts to completely cover a topological manifold.
- ☐ The chart transition map  $y \circ x^{-1}$  between two charts  $(U, x)$  and  $(V, y)$  is continuous.
- ☐ The continuity of a curve in a manifold can be verified with or without the use of charts.
- ☐ Any smooth manifold is also a topological manifold, while the converse holds only for some cases.
- ☐ The atlas of a smooth manifold cannot contain all charts of the underlying topological manifold.

## 2. Mastering the formalism

Mastership is in the basics. Work hard to find the required short proofs. This exercise will pay off.

- (a) Let  $(M, \mathcal{O})$  be a topological space and  $S$  be some subset of  $M$ . Prove that the so-called *subset topology*

$$\mathcal{O}|_S := \{U \cap S \mid U \in \mathcal{O}\}$$

indeed defines a topology on the subset  $S$ .

- (b) Let  $(M, \mathcal{O}_M)$  and  $(N, \mathcal{O}_N)$  be topological spaces. Define the so-called *product topology*  $\mathcal{O}_{M \times N}$  on  $M \times N$  as the set that contains precisely those subsets  $W \subseteq M \times N$  for whose every point  $w = (m, n) \in W$  there exists a  $U \in \mathcal{O}_M$  with  $m \in U$  and a  $V \in \mathcal{O}_N$  with  $n \in V$  such that  $U \times V \subseteq W$ . Show that  $\mathcal{O}_{M \times N}$  indeed defines a topology on the cartesian product set  $M \times N$ . Graphically illustrate the construction for  $M = N = \mathbb{R}$ .
- (c) Show that the so-called *standard topology*  $\mathcal{O}_s$  on  $\mathbb{R}^d$  indeed satisfies the axioms of a topology.
- (d) Prove that the composition  $f \circ \gamma$  of a smooth curve  $\gamma : \mathbb{R} \rightarrow M$  and a smooth function  $f : M \rightarrow \mathbb{R}$  on a smooth manifold  $(M, \mathcal{O}, \mathcal{A})$  is again smooth.
- (e) Let  $(M, \mathcal{O})$  be a topological space. If nothing else is said, how would you interpret the statement that a map

$$\varphi : M \times M \rightarrow \mathbb{R}$$

is continuous?

- (f) Prove that the component maps  $x^1, \dots, x^{\dim M}$  defined by

$$x^a : U \rightarrow \mathbb{R}, \quad p \mapsto a\text{-th entry of the } \dim M\text{-tuple } x(p)$$

for any chart  $(U, x)$  in a smooth atlas  $\mathcal{A}$  of a smooth manifold  $(M, \mathcal{O}, \mathcal{A})$  are smooth functions on  $U$ .